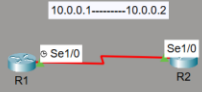


PPP



• 1. PPP (Point-to-Point Protocol)

Question:

Two routers (R1 and R2) are connected via a serial link. Configure PPP encapsulation on both routers. Additionally, ensure the link is operational and verify the configuration.

✦ What you should do:

Configure PPP on both sides
Assign IP addresses
Enable interface
Verify using show commands

💡 Tip (Exam Insight)

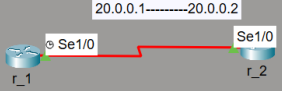
PAP → check username/password manually
CHAP → hostname must match username
PPP → always start with:
encapsulation ppp

```
R1(config)# interface Serial1/0
R1(config-if)# ip address 10.0.0.1 255.255.255.0
R1(config-if)# encapsulation ppp
R1(config-if)# no shutdown

R2(config)# interface Serial1/0
R2(config-if)# ip address 10.0.0.2 255.255.255.0
R2(config-if)# encapsulation ppp
R2(config-if)# no shutdown
```

show ip interface brief
show interfaces serial1/0

PAP



• 2. PAP (Password Authentication Protocol)

Question:

Configure PPP with PAP authentication between two routers (R1 and R2) with the following requirements:

R1 username: R2, password: cisco
R2 username: R1, password: cisco
Use PAP authentication on the serial link

✦ Tasks:

Configure PPP encapsulation
Configure PAP authentication
Set username/password on both routers
Verify authentication success

```
r1(config)#interface Serial1/0
r1(config-if)#ip address 20.0.0.1 255.0.0.0
r1(config-if)#encapsulation ppp
r1(config-if)#ppp authentication pap
r1(config-if)#ppp pap sent-username R1 password cisco
r1(config-if)#no shutdown

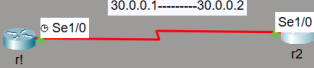
r2(config)#interface Serial1/0
r2(config-if)#ip address 20.0.0.2 255.0.0.0
r2(config-if)#encapsulation ppp
r2(config-if)#ppp authentication pap
r2(config-if)#ppp pap sent-username R2 password cisco
r2(config-if)#no shutdown
r2(config-if)#ex
```

runn conf

shutdown

```
!
interface Serial1/0
ip address 20.0.0.1 255.0.0.0
encapsulation ppp
ppp authentication pap
ppp pap sent-username R1 password 0 cisco
clock rate 2000000
!
```

CHAP



• 3. CHAP (Challenge Handshake Authentication Protocol)

Question:

Configure PPP with CHAP authentication between routers (R1 and R2) with the following conditions:

Hostnames must match usernames
Password on both routers: secure123
Use CHAP authentication on the serial interface

✦ Tasks:

Set correct hostnames
Configure PPP encapsulation
Configure CHAP authentication
Verify using debug/show commands

```
R1(config)# username R2 password secure123
R1(config)#
R1(config)#interface Serial1/0
R1(config-if)#ip address 30.0.0.1 255.0.0.0
R1(config-if)#encapsulation ppp
R1(config-if)# ppp authentication chap
R1(config-if)#no shutdown

R2(config)# username R1 password secure123
R2(config)#
R2(config)#interface Serial1/0
R2(config-if)#ip address 30.0.0.2 255.0.0.0
R2(config-if)#encapsulation ppp
R2(config-if)# ppp authentication chap
R2(config-if)#no shutdown
```

verfy

shutdown

```
!
interface Serial1/0
ip address 30.0.0.2 255.0.0.0
encapsulation ppp
ppp authentication chap
!
interface Serial1/1
no ip address
```